

WIPRO TRAINING CAPSTONE PROJECT

PROJECT NAME : Enterprise UI Automation Testing Framework using Playwright.

PROJECT OVERVIEW : This project is a professional End-to-End UI Automation Testing Framework developed using Playwright for the SauceDemo (Swag Labs) e-commerce application.

The framework is designed to automate and validate critical business workflows including:

- User Authentication
- Product Catalog Validation
- Product Sorting
- Shopping Cart Operations
- Checkout Form Validation
- Order Calculation Verification
- Cross-Browser Execution

The framework follows industry-standard automation practices such as Page Object Model (POM), reusable automation components, parallel execution, assertion-based validation, and automated reporting.

WEBSITE NAME : SauceDemo (Swag Labs)

WEBSITE URL : <https://www.saucedemo.com/>

PROJECT OBJECTIVES :

- Prepare a complete automation test plan for the SauceDemo (Swag Labs) website.
- Create a minimum of 120 automated test cases across core application modules.
- Cover positive, negative, validation, navigation, and regression testing scenarios.
- Automate the planned test cases using Playwright with JavaScript.
- Validate critical workflows such as login, product sorting, cart operations, checkout validation, and order calculations.
- Execute test cases across Chromium, Firefox, and WebKit browsers.

- Generate execution evidence through HTML reports, screenshots, and trace files wherever required.
- Implement automation using the Page Object Model (POM) design pattern and reusable components.
- Present the project as a professional QA Automation Testing Capstone Project.

TEST CASE DISTRIBUTION :

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- Core Test Scenarios : 40
- Browser Configurations : 3
- Total Planned Tests : 120+

PLANNED SERVICES :

1. Authentication & Security

- Valid Login
- Invalid Login
- Session Handling
- Error Message Validation

2. Product Catalog & Sorting

- Product Sorting
- Product Details Validation
- Catalog Verification

3. Cart Operations

- Add Product to Cart
- Remove Product from Cart
- Cart Badge Validation

4. Checkout Validation

- Form Validation

- Empty Field Validation
- Error Message Verification

5. Order Calculations

- Subtotal Verification
- Tax Validation
- Final Total Verification

6. Navigation & UI Validation

- Menu Navigation
- Page Redirection Validation
- Button & Link Verification

7. Product Filtering & Search

- Product Search Validation
- Filter Functionality
- Search Result Verification

8. Cross-Browser Testing

- Chromium Execution
- Firefox Execution
- WebKit Execution

TOOLS & TECHNOLOGIES :

- Playwright
- JavaScript
- Node.js
- Visual Studio Code
- Git & GitHub
- HTML Reporting
- Chromium, Firefox, WebKit

AUTOMATION FRAMEWORK APPROACH :

- Page Object Model (POM)
- Reusable Functions
- Modular Test Structure
- Data-Driven Testing
- Assertion-Based Validation

CONCLUSION :

This project demonstrates a scalable and enterprise-level UI Automation Testing Framework using Playwright for the SauceDemo (Swag Labs) e-commerce application. The framework successfully automates critical user workflows including authentication, product validation, cart operations, checkout validation, and cross-browser testing.